

MH2500: Probability and Intro to Statistics

Tutorial Introduction & Tutorial 1

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About myself

- Tutor's Name: [CHEY Chun Fei](#)
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- Office hours: [Wednesdays 11am-1pm; Thursdays: 3-4pm](#)

Do drop me an email if you require consultation.

About myself

- Graduated from Bachelor of Science in Mathematical Sciences at NTU, Class of 2020, specialised in Applied Mathematics.
- Currently a Second-year PhD student in SPMS from the Division of Mathematical Sciences.
 - Research Supervisor: Assistant Professor Tong Ping.
 - My Research interests (if you are interested to know): [Differential Equations](#), [Computational Mathematics](#) and [Numerical Analysis](#).

Regarding the course: MH2500

- This is an **essential** and **introductory** core module for all Year 2 Mathematics, Data Science & AI students.
- Prerequisites: Calculus I and II or any equivalent Calculus modules.
- Some topics covered: Combinatorics, Discrete and Continuous Distributions, Joint Distributions, Correlation, Hypothesis Testing.
- It is important to know how to apply the theorems and results to solve the problems.
- **Do** as many exercises as possible! Have a discussion with your peers is also useful.

About Tutorials

- Tutorial Classes (Held on Thursdays)
 - **T3: 11.30-12.20, Venue: SPMS-TR+11**
 - **T7: 13.30-14.20, Venue: SPMS-TR+11**
 - **T10: 16.30-17.20, Venue: SPMS-TR+10**
- I will do a short recap of the week's content. ($\approx$ 5-7 mins)
- **Tutorial Presentation (10% of the assessment)** (Starting next week)
 - Present the **tutorial problems** for the week.
 - Write down the solution and explain the steps/thought process to the class. (I may ask questions to the presenter if there are some parts that aren't clear.)
 - Highly encouraged to ask questions or you may present an alternative method.
 - If time permits, some assignment problems will be discussed.

Definition

The set of all possible outcomes of a statistical experiment is called the **sample space** and is represented by the symbol S , or sometimes Ω .

Theorem (Extended Multiplicative Principle)

If there are p experiments with the first experiment having n_1 outcomes, the second experiment having n_2 outcomes, $\dots$, the p -th experiment having n_p outcomes, then there are $n_1 n_2 \cdots n_p$ possible outcomes for the p experiments.

Theorem (Permutations)

- ① *The number of permutations of n distinct objects is $n!$.*
- ② *The number of permutations of n distinct objects taken r at a time is*

$${}_nP_r = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!} \text{ or } \binom{n}{r} \times r!$$

Theorem (Arrangements with Repetition)

The number of distinct permutations of n objects, of which n_1 are of one kind, n_2 of a second, $\dots$, n_k of a k -th kind, is

$$\frac{n!}{n_1!n_2!\dots n_k!} \text{ denoted as } \binom{n}{n_1, n_2, \dots, n_k}.$$

For a set of size n and a sample of size r , there are n^r different ordered samples with replacement and ${}_nP_r$ different ordered samples without replacement.

Theorem (Combinations)

The number of combinations of n distinct objects, taken r at a time is

$$\binom{n}{r} = \frac{n!}{(n-r)!r!}.$$

A **permutation** is an ordered arrangement of objects.

A **combination** is a collection of objects where ordering does not matter.

Please help me to fill in this google form (QR code or tiny URL) to allow me to communicate with you better via e-mail and share weekly summary slides as well. Thanks!



tinyurl.com/tut2500

[Link to tutorial slides and handwritten solution to tutorial 1:](https://shorturl.at/vxEGR)

shorturl.at/vxEGR

[Link to recording of Tutorial 1:](https://ntu-sg.zoom.us/rec/share/tpy1xp62LC4Zz98J4ADZ-d2YRgyf6ZCQmclWtpdVEC92zjyBBavVIvBc60MDcjEq.nwcl6kFxhnDftNBC)

<https://ntu-sg.zoom.us/rec/share/tpy1xp62LC4Zz98J4ADZ-d2YRgyf6ZCQmclWtpdVEC92zjyBBavVIvBc60MDcjEq.nwcl6kFxhnDftNBC>

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